1 FIG-002 — Mace 1ns evidence aware benchmark

Group: MD-1NS **Evidence chain:** manifested **Status:** publication

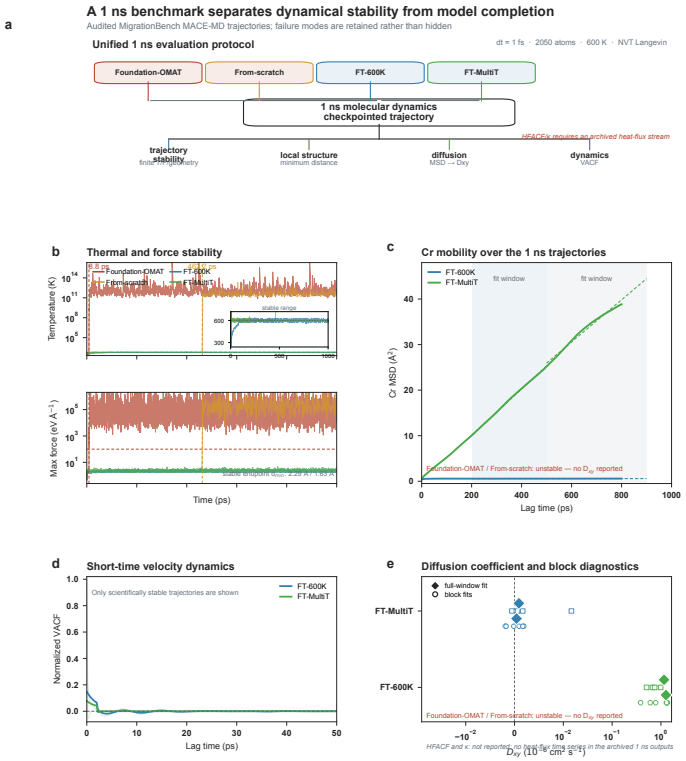


Figure 1: One-nanosecond molecular-dynamics stability and diffusion diagnostics across the audited model set. This page shows the candidate output “Mace 1ns evidence aware benchmark” within the audited MD-1NS family. The panel identity, source scope, and current status are retained in the accompanying evidence matrix.

Results text draft. The panels summarize trajectory integrity, thermal/force behavior, structural diagnostics, and diffusion observables where the required trajectory and convergence gates are available. Failed or excluded branches remain visible rather than being converted into numerical transport estimates. **Discussion and boundary.** This figure can support dynamic-stability discussion. It must not be used to imply heat-flux-derived thermal conductivity when the same-campaign heat-flux series is absent; such quantities remain explicitly unavailable. **Article placement.** Proposed article location: Results: dynamic stability and diffusion; SI transport candidate. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Resolve the asymmetric FT-600K MD transport protocol and its effect on D and kappa. *Completed:* Legacy FT-600K transport values are quarantined and excluded from ranking; the figure can document that boundary. *Still required:* Run the corrected matched-protocol FT-600K long trajectory, block-average D/kappa, and close the multi-seed uncertainty before reporting a final transport comparison.

Evidence pointers.

- Data/processed: `main/analysis/mace_1ns_publication_figure_20261005/figure_manifest.json`
- Data/processed: `main/analysis/mace_1ns_publication_figure_20261005/final_benchmark_manifest.json`
- Plot/processing script: `main/analysis/mace_1ns_publication_figure_20261005/plot_figure.py`
- Plot/processing script: `main/analysis/mace_1ns_publication_figure_20261005/plot_final_benchmark.py`
- Caption/text source: `main/analysis/mace_1ns_publication_figure_20261005/figure_caption.md`
- Caption/text source: `main/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.md`

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/figure_caption.md
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.md
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_caption.tex
- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/strict_replica_caption.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/figure_manifest.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/final_benchmark_manifest.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/overleaf_panels_manifest.json
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- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_final_benchmark.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_overleaf_panels.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/plot_strict_replica.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_lag_remote.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_msd_fast.py
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_publication_figure_20261005/recompute_full_vacf_fast.py

Audit gap / next action: Article integration and heat-flux absence must remain explicit